

Concepts of Security Zoning:

- o In Cisco ASA Firewall Security Zones are the building blocks for policies.
- o They are logical entities to which one or more interfaces are bound.
- o Defining Zones is part of Security framework which needs to be followed.
- o Without Zones, required level of security across assets may not be possible.
- o In ASA, Security Zones provide a means of distinguishing groups of hosts.
- o Security Level is assigned to interfaces either physical or logical sub-interfaces.
- o It is basically a number from 0 to 100 designating how trusted an interface is.
- o Cisco ASA firewall can be configured to have multiple security levels from 0 to 100.
- o Security Level indicates how trusted an interface is compared to other interfaces.
- o The higher the security level, the more trusted the physical or logical interface.
- o Since each ASA firewall interface represents a specific network or security zone.
- o The Traffic from a high security level is permitted to go to a lower security level.
- o Traffic from low security level to higher security level will be denied, until allowed.

Inside Zone:

- o This is the local LAN (Local Area Network) side of the Cisco ASA Firewall.
- o Typical high security, #100 and trusted, it is called Private and Trusted Zone.
- o This is the highest and most trusted security level of Cisco ASA Firewall.

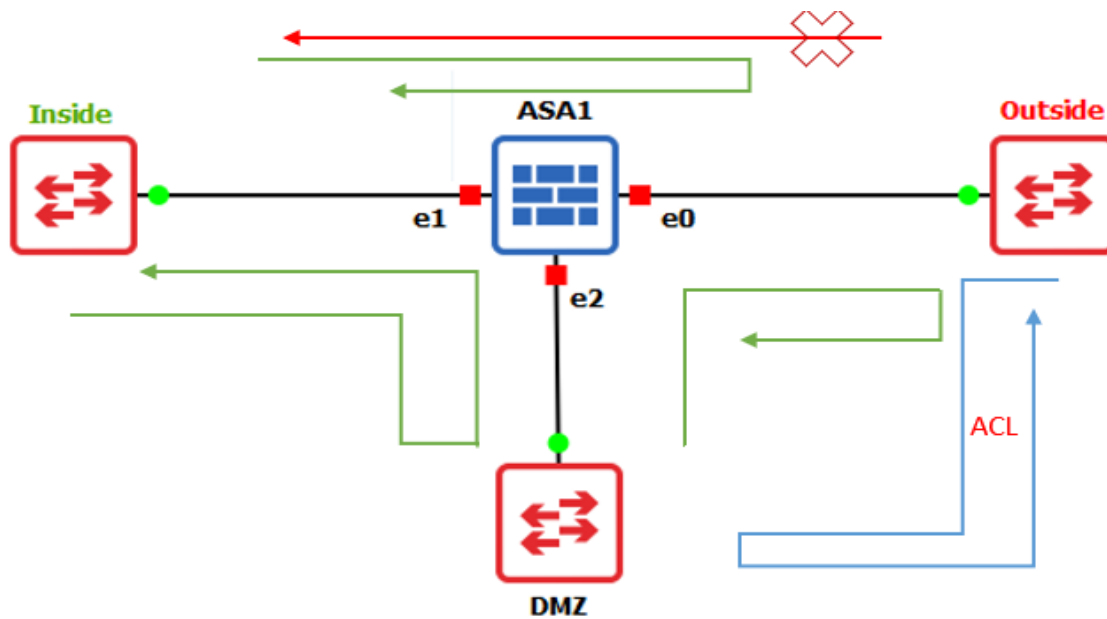
Outside Zone:

- o Outside Security Zone is assign to the Internet or Public Network.
- o Outside zone typically assign lowest security #0, not trusted at all.
- o Outside Security Zone is also called Public Zone and Untrusted Zone.
- o This is lowest & considered least secured Security Level on ASA Firewall.

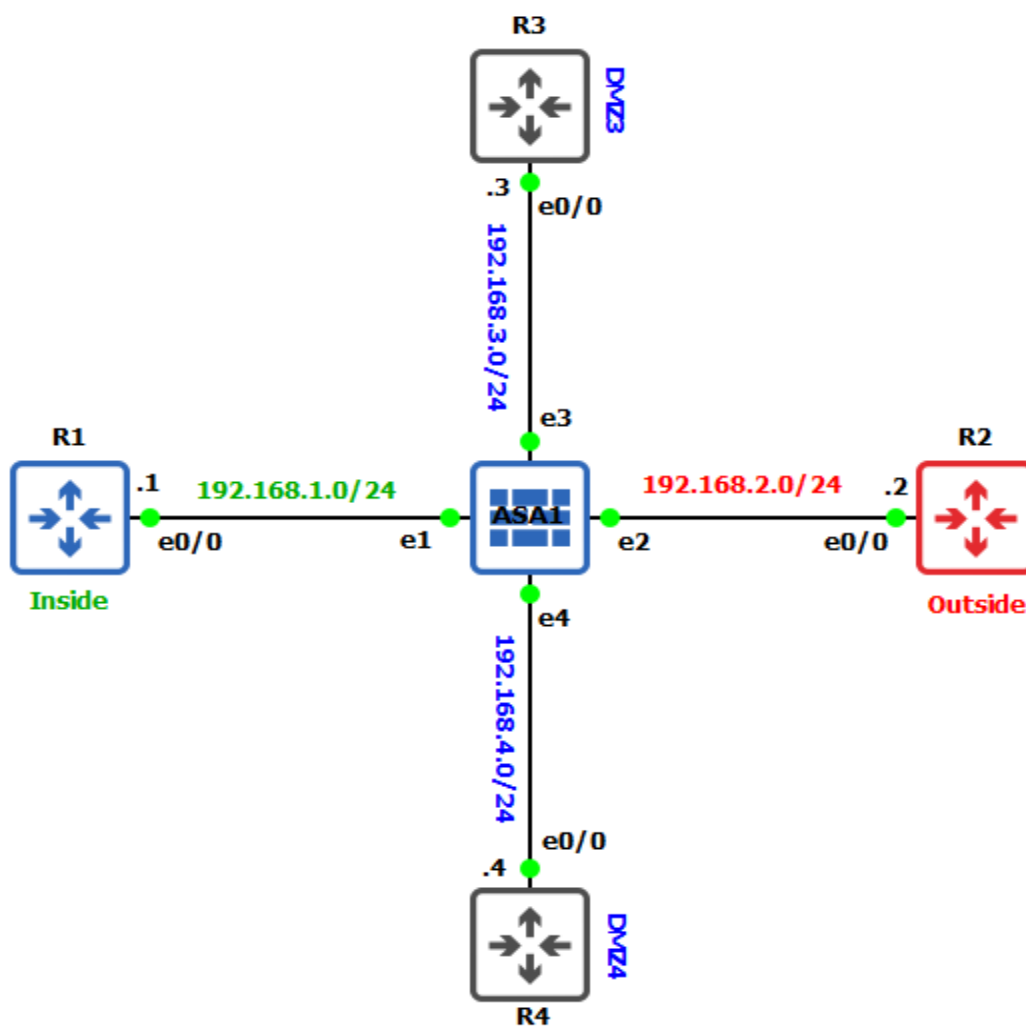
DMZ:

- o DMZ (De-Militarized Zone) Zone is a portion of your company network.
- o Which, although under your control, is outside your heaviest security.
- o Compared to rest of network, machines place in DMZ are less protected.
- o This DMZ (De-Militarized Zone) has one foot in each camp or in zone.
- o Normally DMZ zone, typically assign security number from 1 to 99 number.
- o Traffic from a "High" security level to a "Lower" security level is permitted.
- o Traffic from a "Low" security level to a "Higher" security level is denied.
- o Traffic between two interfaces of the same security level is also dropped.

Traffic from INSIDE to OUTSIDE, is permitted	Traffic from DMZ to INSIDE, is denied
Traffic from INSIDE to DMZ, is permitted	Traffic from OUTSIDE to DMZ, is denied
Traffic from DMZ to OUTSIDE, is permitted	Traffic from OUTSIDE to INSIDE, is denied



Lab Time:



Routers Configuration

```
R1(config)# interface Ethernet0/0
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.254
```

```
R2(config)# interface Ethernet0/0
R2(config-if)#ip address 192.168.2.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)# ip route 0.0.0.0 0.0.0.0 192.168.2.254
```

```
R3(config)# interface Ethernet0/0
R3(config-if)#ip address 192.168.3.3 255.255.255.0
R3(config-if)#no shutdown
R3(config)# ip route 0.0.0.0 0.0.0.0 192.168.3.254
```

```
R4(config)# interface Ethernet0/0
R4(config-if)#ip address 192.168.4.4 255.255.255.0
R4(config-if)#no shutdown
R4(config)# ip route 0.0.0.0 0.0.0.0 192.168.4.254
```

ASA1 Configuration

```
ASA1(config)#interface Ethernet1
ASA1(config-if)#nameif inside
ASA1(config-if)#security-level 100
ASA1(config-if)#ip address 192.168.1.254
ASA1(config-if)#no shutdown
```

```
ASA1(config)#interface Ethernet2
ASA1(config-if)#nameif outside
ASA1(config-if)#security-level 0
ASA1(config-if)#ip address 192.168.2.254
ASA1(config-if)#no shutdown
```

```
ASA1(config)#interface Ethernet3
ASA1(config-if)#nameif DMZ3
ASA1(config-if)#security-level 50
ASA1(config-if)#ip address 192.168.3.254
ASA1(config-if)#no shutdown
```

```
ASA1(config)#interface Ethernet4
ASA1(config-if)#nameif DMZ4
ASA1(config-if)#security-level 50
ASA1(config-if)#ip address 192.168.4.254
ASA1(config-if)#no shutdown
```

```
ASA1(config)#fixup protocol icmp
```

```
ASA1(config)#same-security-traffic permit inter-interface
```

Verification:

- 1- Ping from R1 to R2, R3 and R4 are **successful**, Traffic from High security level to Lower level.
- 2- Ping from R2 to R1, R3 and R4 are **unsuccessful**, Traffic from Low security level to a Higher.
- 3- Ping from R3 to R1 are **unsuccessful**, Traffic from Low security level to Higher security level.
- 4- Ping from R3 to R2 **successful**, Traffic from High security level to Lower security level.
- 5- Ping from R3 to R4 **unsuccessful**, Traffic between two interfaces of the same security level.
- 6- Ping from R4 to R1 are **unsuccessful**, Traffic from Low security level to Higher security level.
- 7- Ping from R4 to R2 **successful**, Traffic from High security level to Lower security level.
- 8- Ping from R4 to R3 **unsuccessful**, Traffic between two interfaces of the same security level.
- 9- After apply this command on cisco ASA firewall, same-security-traffic permit inter-interface, Ping from R3 to R4 and R4 to R3 **successful**.

Source	Destination	Result
Inside	DMZ	Pass
Inside	Outside	Pass
DMZ	Inside	Drop
DMZ	Outside	Pass
DMZ	DMZ	Drop
Outside	Inside	Drop
Outside	DMZ	Drop

Ping Test from R1
R1#ping 192.168.2.2
R1#ping 192.168.3.3
R1#ping 192.168.4.4
Ping Test from R2
R2#ping 192.168.1.1
R2#ping 192.168.3.3
R2#ping 192.168.4.4
Ping Test from R3
R3#ping 192.168.1.1
R3#ping 192.168.2.2
R3#ping 192.168.4.4
Ping Test from R4
R4#ping 192.168.1.1
R4#ping 192.168.2.2
R4#ping 192.168.3.3